



RIPHAH
INTERNATIONAL UNIVERSITY

BS-AI

LAB TASK 10

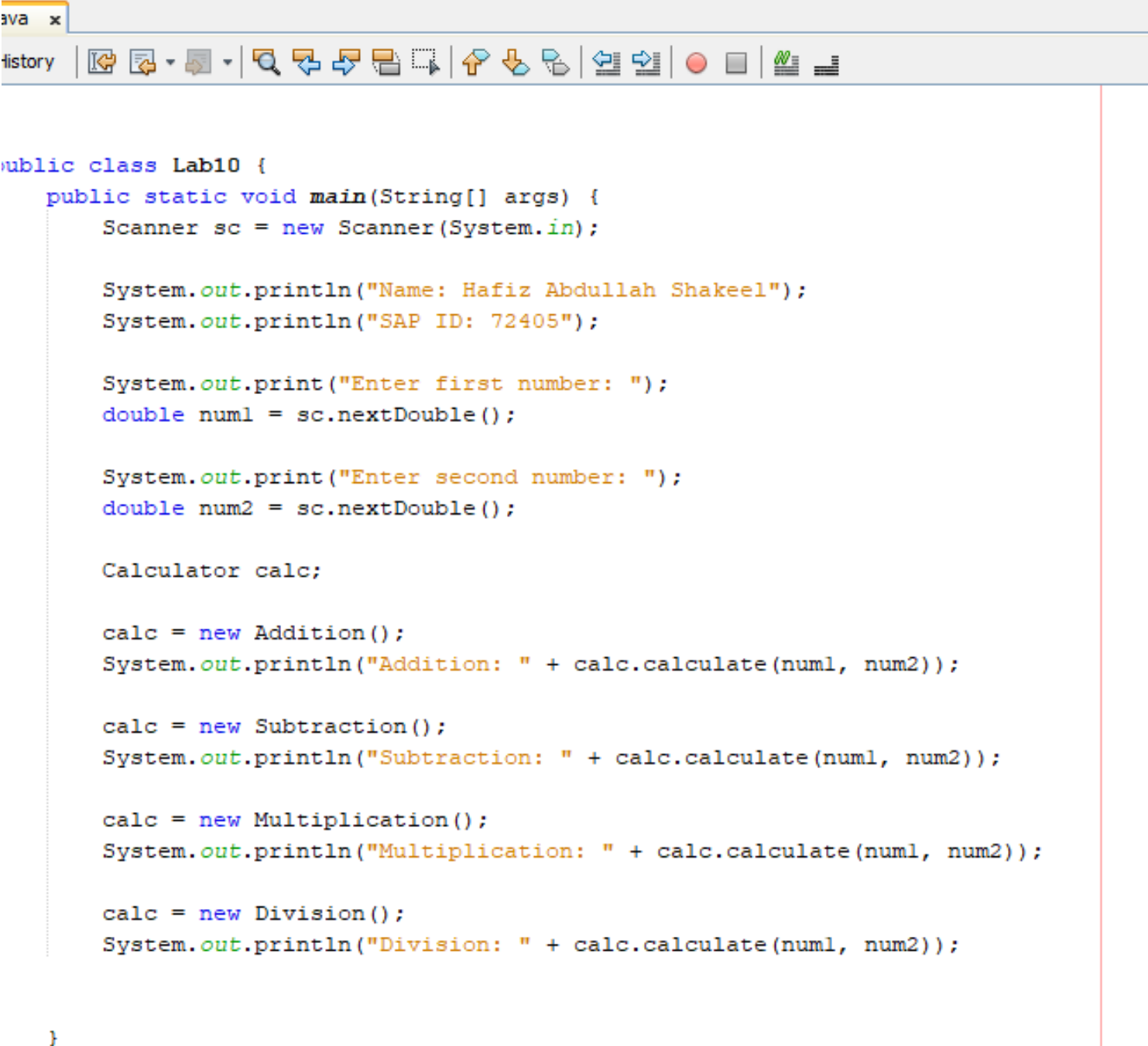
Name: Hafiz Abdullah Shakeel

Sap I'D: 72405

Subject: Object Orientated
Programming

Teacher: Sir Imran Khan

Task 1:



```
public class Lab10 {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("Name: Hafiz Abdullah Shakeel");  
        System.out.println("SAP ID: 72405");  
  
        System.out.print("Enter first number: ");  
        double num1 = sc.nextDouble();  
  
        System.out.print("Enter second number: ");  
        double num2 = sc.nextDouble();  
  
        Calculator calc;  
  
        calc = new Addition();  
        System.out.println("Addition: " + calc.calculate(num1, num2));  
  
        calc = new Subtraction();  
        System.out.println("Subtraction: " + calc.calculate(num1, num2));  
  
        calc = new Multiplication();  
        System.out.println("Multiplication: " + calc.calculate(num1, num2));  
  
        calc = new Division();  
        System.out.println("Division: " + calc.calculate(num1, num2));  
    }  
}
```

Lab10.java x

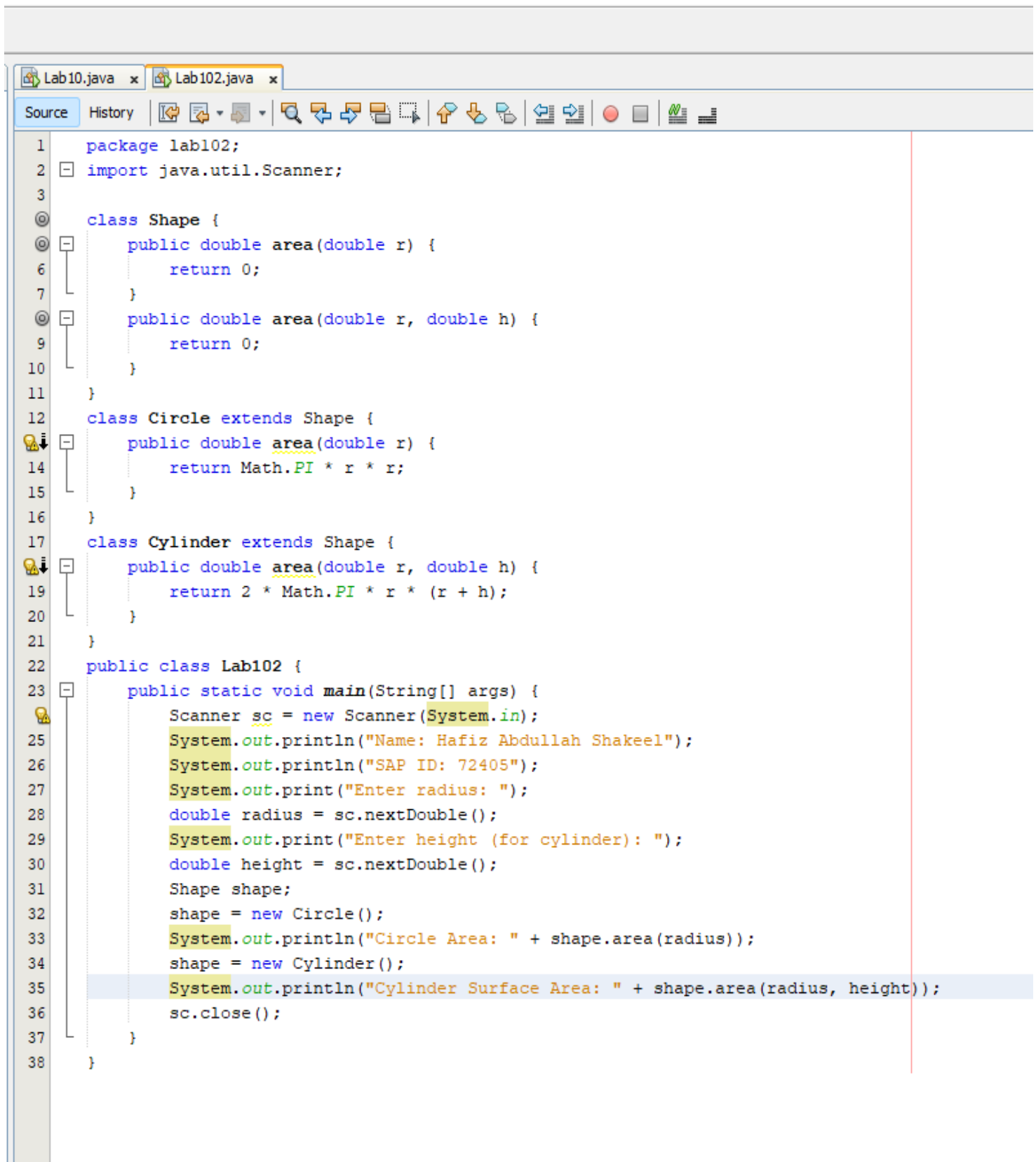
SourceHistory

```
1 package lab10;
2 import java.util.Scanner;
3
4 class Calculator {
5     public double calculate(double a, double b) {
6         return a + b;
7     }
8 }
9
10 class Addition extends Calculator {
11     public double calculate(double a, double b) {
12         return a + b;
13     }
14 }
15
16 class Subtraction extends Calculator {
17     public double calculate(double a, double b) {
18         return a - b;
19     }
20 }
21
22 class Multiplication extends Calculator {
23     public double calculate(double a, double b) {
24         return a * b;
25     }
26 }
27
28 class Division extends Calculator {
29     public double calculate(double a, double b) {
30         if (b != 0) {
31             return a / b;
32         } else {
33             System.out.println("Cannot divide by zero!");
34             return 0;
35         }
36     }
37 }
38
39 public class Lab10 {
40     public static void main(String[] args) {
41         Scanner sc = new Scanner(System.in);
42
43         System.out.println("Name: Hafiz Abdullah Shakeel");
44         System.out.println("SAP ID: 72405");
45
46         System.out.print("Enter first number: ");
```

Output - lab10 (run) x

>> run:

Task 2:



```
1 package lab102;
2 import java.util.Scanner;
3
4 class Shape {
5     public double area(double r) {
6         return 0;
7     }
8     public double area(double r, double h) {
9         return 0;
10    }
11 }
12 class Circle extends Shape {
13     public double area(double r) {
14         return Math.PI * r * r;
15     }
16 }
17 class Cylinder extends Shape {
18     public double area(double r, double h) {
19         return 2 * Math.PI * r * (r + h);
20     }
21 }
22 public class Lab102 {
23     public static void main(String[] args) {
24         Scanner sc = new Scanner(System.in);
25         System.out.println("Name: Hafiz Abdullah Shakeel");
26         System.out.println("SAP ID: 72405");
27         System.out.print("Enter radius: ");
28         double radius = sc.nextDouble();
29         System.out.print("Enter height (for cylinder): ");
30         double height = sc.nextDouble();
31         Shape shape;
32         shape = new Circle();
33         System.out.println("Circle Area: " + shape.area(radius));
34         shape = new Cylinder();
35         System.out.println("Cylinder Surface Area: " + shape.area(radius, height));
36         sc.close();
37     }
38 }
```